STEM Course Qualitative Evaluation Analysis

Thematic Analysis of Student Course Evaluation Comments

Dataset	STEM_Course_Qual_Eval.xlsx
Total Rows	7,989 records
Analysable Comments	7,760 (after removing blanks)
Unique Comments	6,975 (785 duplicates removed)
Time Span	Fall 2013 – Spring 2024 (22 terms)
Comment Types	General (7,406) / Cheating (583)
Analysis Method	Inductive Thematic Analysis + Keyword Coding
Date of Analysis	March 2026

1. Executive Summary

Teaching Quality is overwhelmingly the most salient theme across the dataset, appearing in **~28%** of all unique comments (n ≈ 1,926). Students mention instructor characteristics such as engagement, clarity, caring, and knowledge far more frequently than any structural or administrative concern. This signals that the instructor relationship is the primary driver of both satisfaction and dissatisfaction in STEM courses at this institution.

Course Content is the second most prominent theme (n ≈ 850), driven largely by positive characterisations of the material as interesting or enjoyable, but tempered by recurring concerns about difficulty and, less frequently, relevance.

Communication (n ≈ 526) and **Student Engagement** (n ≈ 470) rank third and fourth. The communication theme encompasses instructor responsiveness and office hours; the engagement theme largely reflects students commenting on participatory, discussion-based formats.

Assessment and Exam concerns (n ≈ 454) represent the most concentrated pain point: students report exams as difficult, misaligned with lecture content, and occasionally unfairly graded. These concerns intensify in Chemistry and Psychology courses.

Workload (n ≈ 325), **Course Organisation** (n ≈ 304), and **Recitations & TAs** (n ≈ 294) round out the main themes. Busy-work concerns and disorganisation are notable negative sub-themes.

Cheating (n = 583 dedicated comments) is dominated by **clicker fraud** (n ≈ 57) and **peer copying during exams** (n ≈ 47). ChatGPT/AI misuse is an emerging concern in the most recent terms. Notably, approximately 46 cheating-comment respondents reported observing **no cheating**, providing useful baseline context.

Key Finding: Instructor Impact is the Central Driver
The single strongest signal in the STEM course evaluation dataset is the association between instructor quality and student satisfaction. Comments citing engaging, knowledgeable, or caring instructors are consistently paired with positive course evaluations, while poor explanation and disorganisation from instructors are consistently paired with frustration. Structural interventions (better Canvas pages, clearer rubrics) matter but are secondary to instructor performance.

2. Dataset Overview

2.1 Volume and Composition

Metric	Count	Notes
Total records	7,989	Including all rows in source file
Blank comments	229	Excluded from analysis
Analysable comments	7,760	After removing blanks
Duplicate comments	785	Identical text; first occurrence retained
Unique comments	6,975	Final analytical corpus
General Course Comments	7,406 (92.7%)	
Cheating Comments	583 (7.3%)	
Very short (<20 chars)	344	Retained but flagged; low analytical value

2.2 Time Coverage

The dataset spans 22 academic terms from Fall 2013 through Spring 2024, with the heaviest concentration in Fall 2022 (n=1,019), Fall 2023 (n=1,130), and Spring 2024 (n=997). Earlier terms (2013–2017) have smaller sample sizes and should be interpreted cautiously in trend analyses.

2.3 Course and Instructor Coverage

The most frequently evaluated course is **PSYC001** (n=1,630), followed by **LING001** (n=634), **CHEM1022/CHEM1012** (~328 each), and **ENVS100** (n=258). The most evaluated instructors are **CONNOLLY, CAROLINE J.** (n=492), **WARD, ANDREW H** (n=435), **JENKINS, ADRIANNA C** (n=299), **PLANTE, ALAIN** (n=266), and **SWINGLEY, DANIEL** (n=265).

2.4 Sector Attributes

Comments are distributed across three sector attributes: AULW (n=2,907 — 36%), AUNM (n=2,608 — 33%), and AUPW (n=2,245 — 28%). These are consistent in their thematic profiles and are not analysed separately in detail.

3. Analytical Method

3.1 Qualitative Approach

This analysis follows an inductive thematic analysis framework. The approach prioritises the student voice and resists forcing pre-determined categories onto the data. The process proceeded in the following stages:

1. Data Familiarisation: A stratified random sample of ~200 comments was read closely before any coding commenced, to develop a grounded understanding of the language, tone, and concerns students use.
2. Data Cleaning: Blank and duplicate comments were identified and removed. Comments shorter than 20 characters were retained for frequency counts but given reduced weight in interpretive analysis. Personally identifiable elements (instructor Penn IDs, section codes) were not surfaced in outputs.
3. Open Coding: An inductive coding pass was conducted, generating 41 initial codes drawn from recurring language patterns in the comments. Codes were defined explicitly and tested against edge cases.
4. Code Consolidation: Similar codes were merged. For example, 'does not answer questions well' and 'unclear explanation' were both captured under the poor_explanation code. The final codebook contains 35 refined codes.
5. Thematic Development: Codes were clustered into 9 major themes based on conceptual proximity. Codes were applied using pattern matching and keyword regular expressions to allow quantification of theme prevalence.
6. Comparative Analysis: Themes were compared across Comment Type (General vs. Cheating), Derived Term (longitudinal), Course ID (top 8 courses by volume), and Instructor Name (top 8 by volume).

3.2 Quantification Note

Frequency counts represent the number of unique comments in which a given theme or code is present and not word frequency. A comment may be coded for multiple themes simultaneously. Percentages are proportions of the 6,975 unique analysable comments unless otherwise stated.

4. Codebook

The codebook below presents a selected set of the most analytically important codes, their definitions, inclusion/exclusion criteria, example quotes, and theme assignments.

Code	Definition	Inclusion Criteria	Example Quote	Theme
engaging_lecturer	Instructor makes lectures stimulating	Engag*, enthusiast*, captivat*, passion*	*"His enthusiasm and passion for the subject made each class truly enjoyable."*	Teaching Quality
poor_explanation	Instructor's explanations are unclear	confus*, unclear, hard to follow/understand	*"Lectures were confusing and hard to follow."*	Teaching Quality
instructor_caring	Instructor shows genuine concern for students	approachable, supportive, kind, caring about students	*"She genuinely cares about her students and makes herself available."*	Teaching Quality
reading_off_slides	Instructor reads directly from slides	'reads off slides', 'reads from screen', 'just read'	*"The professor just read from slides with little added insight."*	Teaching Quality
disorganized	Course structure is unclear or chaotic	disorganiz*, unorganized, no structure, chaotic	*"Very poorly organized; required lots of work, yet I learned nothing."*	Course Organisation
canvas_issues	LMS (Canvas) is unhelpful, outdated, or confusing	Mentions of Canvas, course portal issues	*"Canvas was completely out of date — hard to know what was due."*	Course Organisation
exam_difficult	Exams are harder than	exam/test + hard/difficult/tough/challenging	*"The exam difficulty was incomparable*	Assessment & Exams

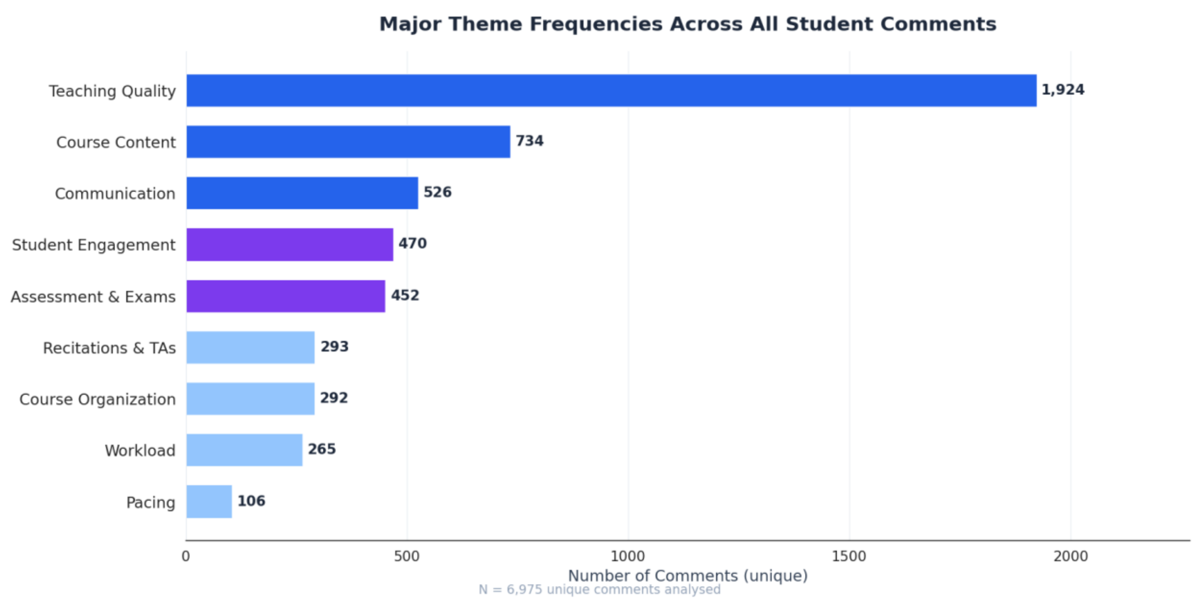
	expected or appropriate		*to homework and practice exams."*	
exam_mismatch	Exam content diverges from lecture coverage	exam deviates from, not covered in lecture, different from slides	*"Test questions were very different from what was covered in lecture."*	Assessment & Exams
lack_feedback	Insufficient or unhelpful feedback on work	no/lack of/not enough feedback	*"We never received feedback on assignments; impossible to improve."*	Assessment & Exams
heavy_workload	Assignment volume is excessive	too much work, heavy workload, overwhelm*	*"Work was overwhelming between midterms."*	Workload
busy_work	Tasks are tedious and add little learning value	busy work, busywork, tedious, pointless assignment	*"Recitation assignments were tedious and did not further knowledge."*	Workload
responsive_instructor	Instructor responds promptly and is easy to reach	prompt*, responsive, easy to reach/contact, available	*"Professor always responded quickly and was easy to approach."*	Communication
poor_communication	Instructor is difficult to reach or unresponsive	unresponsive, no response, hard to reach, poor communication	*"It took weeks to get a response to a simple email."*	Communication
interesting_content	Student finds the subject matter engaging	interesting course/material/content, fascinating	*"The material was fascinating and unlike anything I had studied before."*	Course Content
unhelpful_recitation	Recitation sections add little value	unhelpful/useless/waste of time + recitation/section/TA	*"Recitations were a waste of time and graded too harshly."*	Recitations & TAs

ta_inconsistency	TAs apply rules inconsistently across sections	TA + inconsistent/different/vary	*"Some TAs allowed late arrivals; others did not."*	Recitations & TAs

5. Visualizations

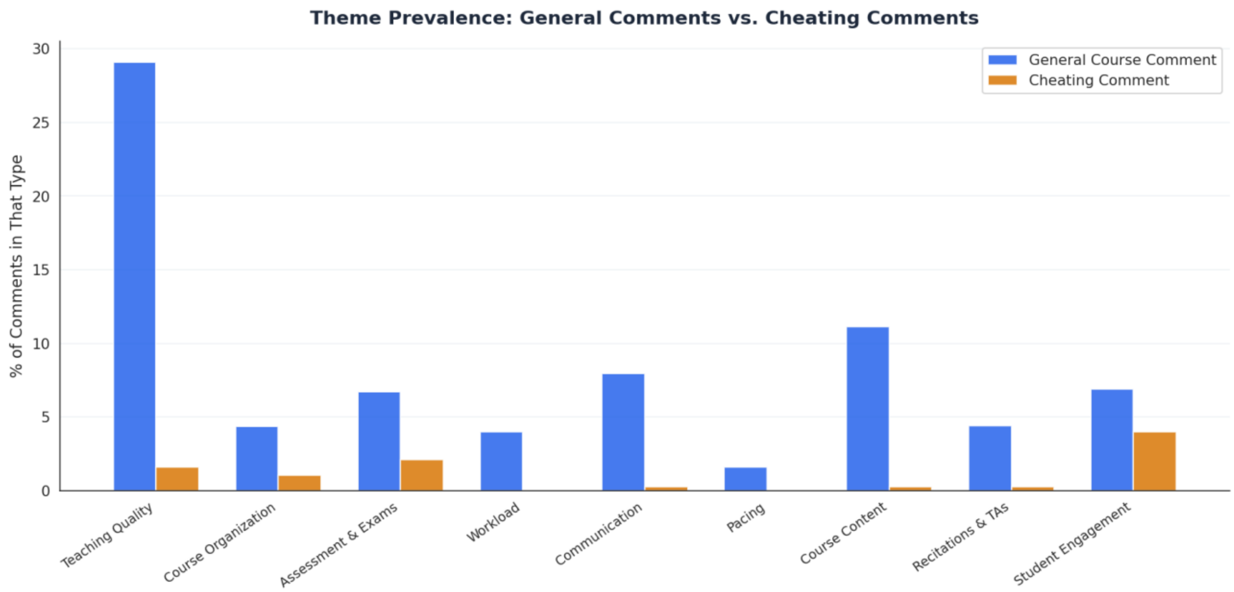
5.1 Major Theme Frequencies

The chart below shows how often each major theme appears across all 6,975 unique comments. Teaching Quality dominates, followed by Course Content and Communication.



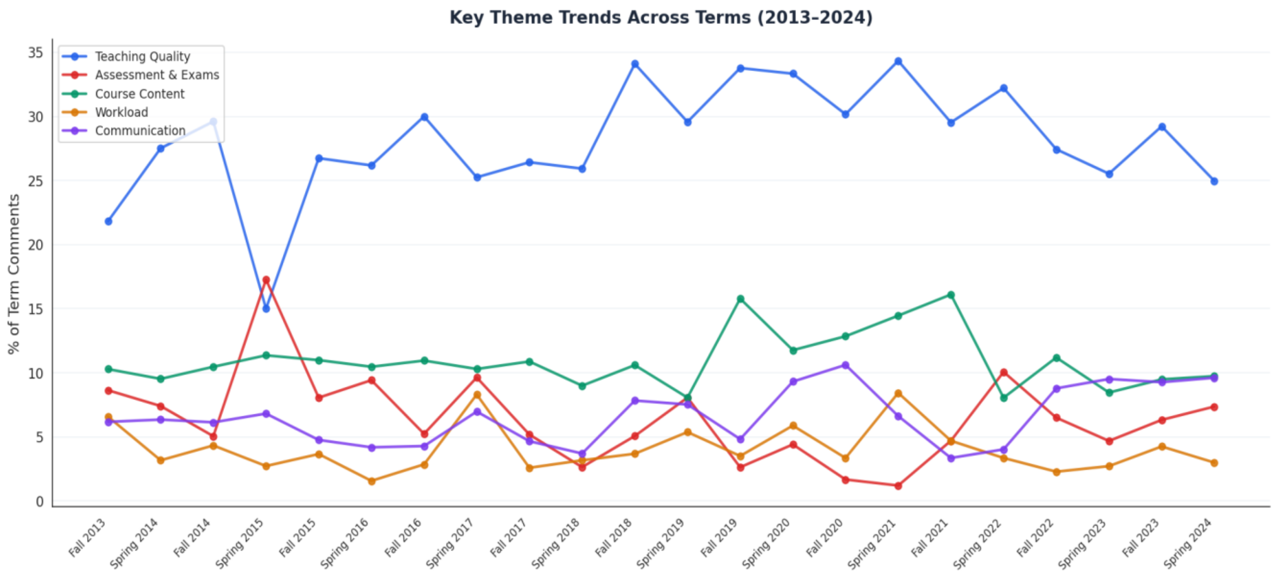
5.2 Theme Comparison: General vs. Cheating Comments

The stacked bar chart compares theme prevalence as a percentage of comments within each comment type. Cheating comments show a distinct profile: Communication and Teaching Quality themes are less prominent, while Course Organisation and Assessment themes appear at comparable rates, likely because students contextualise cheating incidents within course structure.


Theme Prevalence: General Comments vs. Cheating Comments
General Course Comment
Cheating Comment
% of Comments in That Type
0
5
10
15
20
25
30
Teaching Quality
Course Organization
Assessment & Exams
Workload
Communication
Pacing
Course Content
Recitations & TAs
Student Engagement

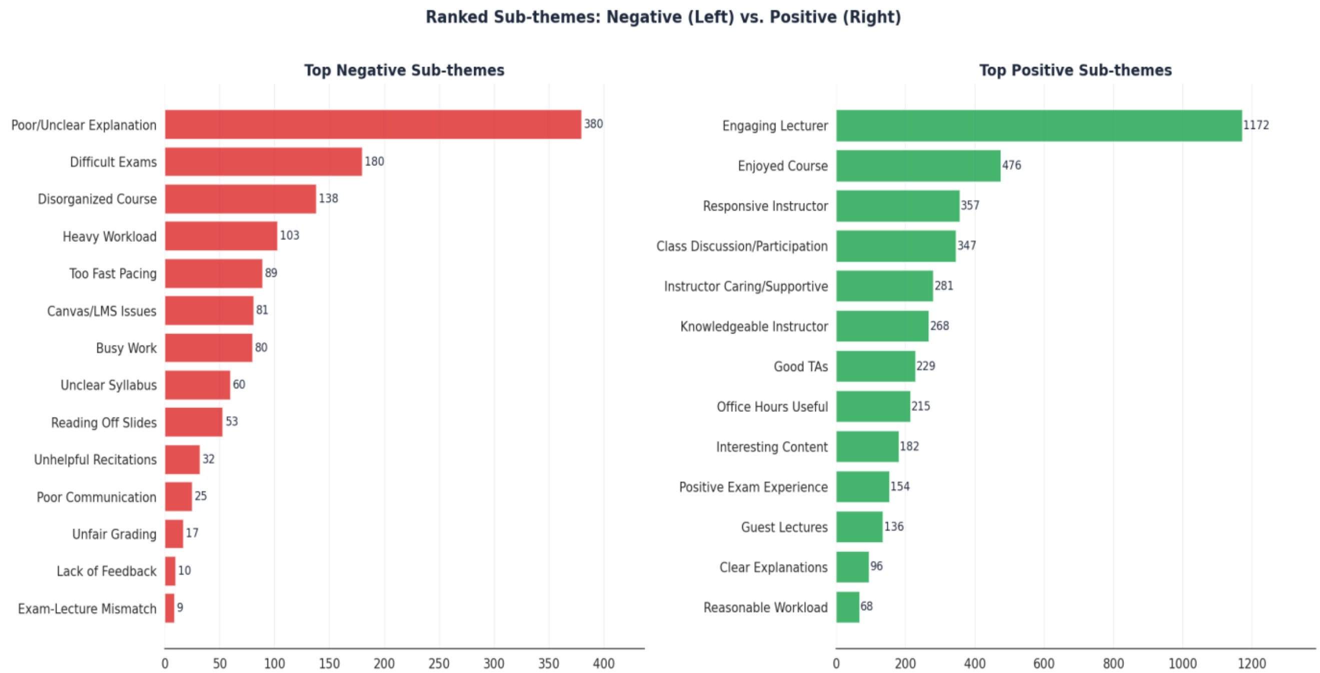
5.3 Theme Trends Across Terms (2013–2024)

The trend chart tracks five key themes over 22 academic terms. Teaching Quality remains the consistently dominant theme. Assessment & Exams concerns spiked in Spring 2015 and Spring 2022. Communication mentions increased noticeably from 2019–2024, possibly reflecting a shift in student expectations for instructor accessibility.



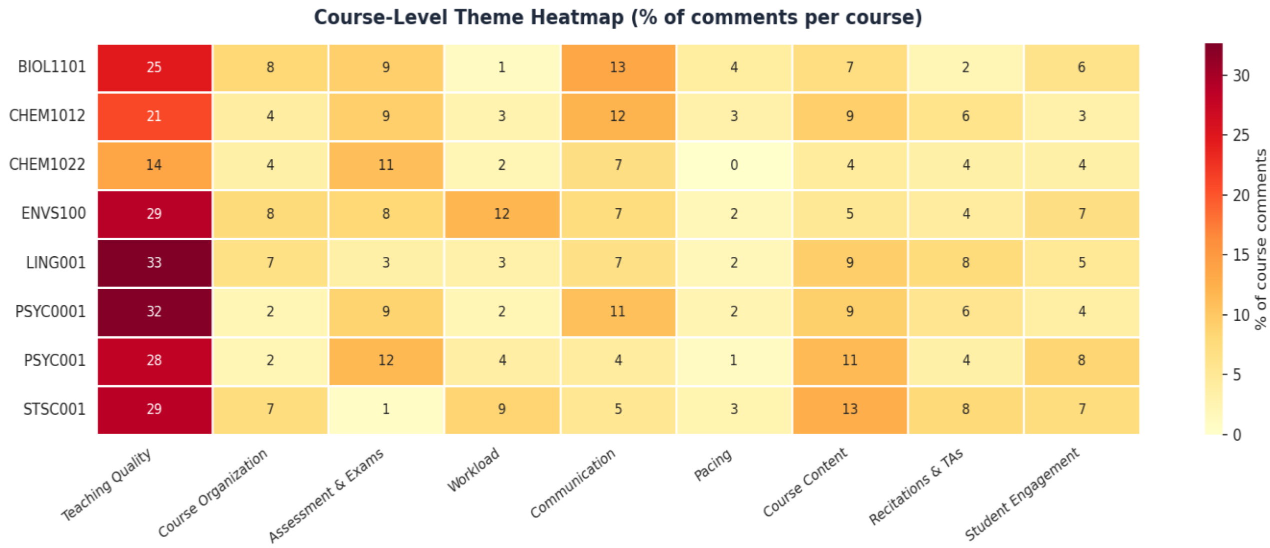
5.4 Positive and Negative Sub-theme Rankings

The panel below presents ranked sub-themes, separated into negative concerns (left) and positive strengths (right). Engaging Lecturer is by far the most common positive signal. Poor/Unclear Explanation is the leading negative sub-theme, followed by Difficult Exams and Disorganized Course.


Ranked Sub-themes: Negative (Left) vs. Positive (Right)
Top Negative Sub-themes
Poor/Unclear Explanation 380
Difficult Exams 180
Disorganized Course 138
Heavy Workload 103
Too Fast Pacing 89
Canvas/LMS Issues 81
Busy Work 80
Unclear Syllabus 60
Reading Off Slides 53
Unhelpful Recitations 32
Poor Communication 25
Unfair Grading 17
Lack of Feedback 10
Exam-Lecture Mismatch 9
0 50 100 150 200 250 300 350 400
Top Positive Sub-themes
Engaging Lecturer 1172
Enjoyed Course 476
Responsive Instructor 357
Class Discussion/Participation 347
Instructor Caring/Supportive 281
Knowledgeable Instructor 268
Good TAs 229
Office Hours Useful 215
Interesting Content 182
Positive Exam Experience 154
Guest Lectures 136
Clear Explanations 96
Reasonable Workload 68
0 200 400 600 800 1000 1200

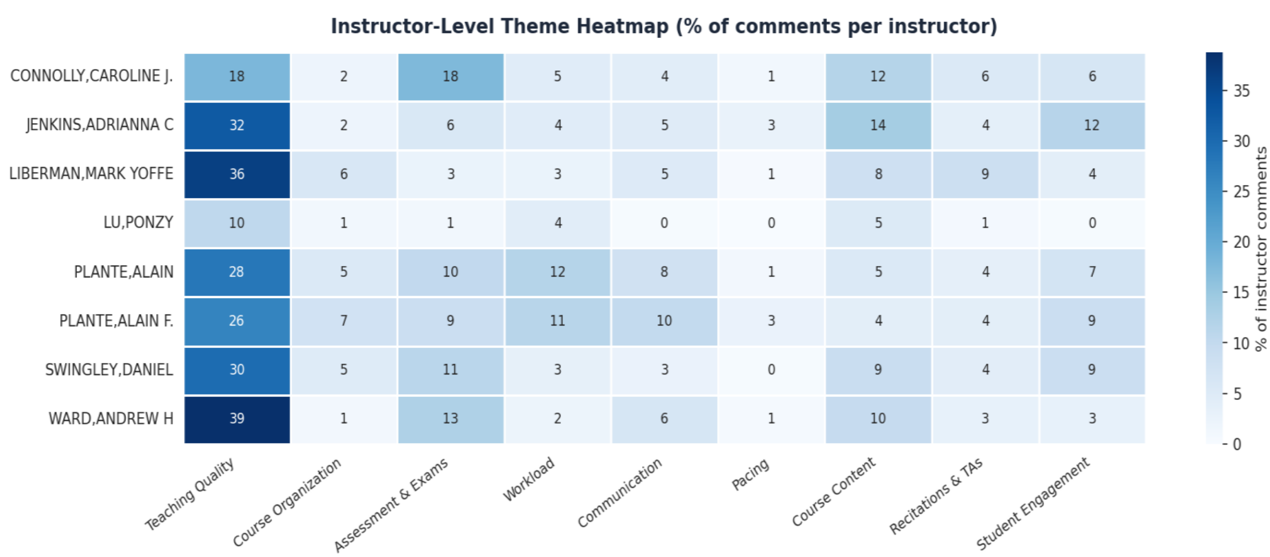
5.5 Course-Level Theme Heatmap

The heatmap shows theme density (% of comments) across the eight highest-volume courses. ENVS100 and STSC001 show notably higher Workload and Course Content scores respectively. CHEM1022 and PSYC001 show elevated Assessment & Exams scores. LING001 and PSYC0001 show the strongest Teaching Quality prominence.



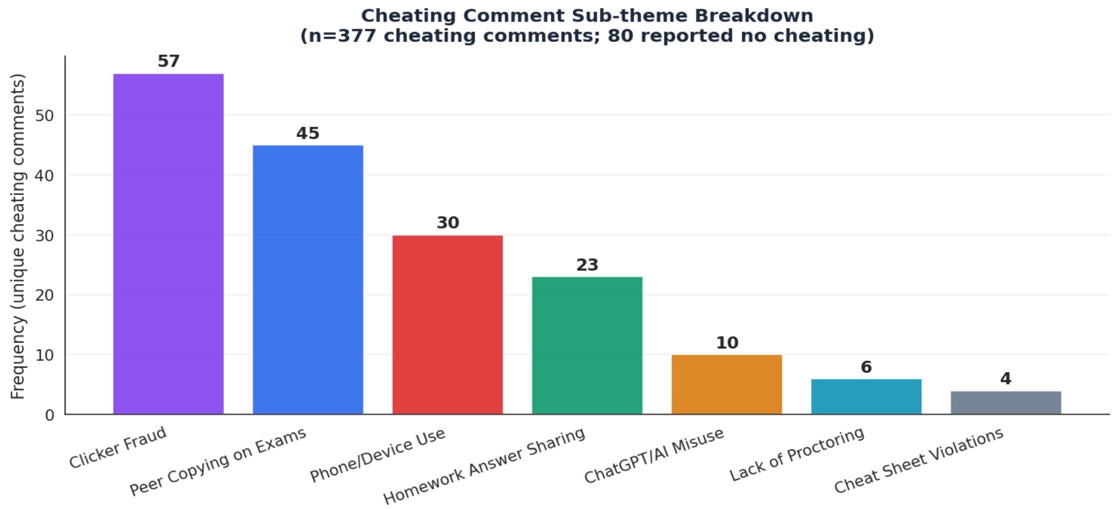
5.6 Instructor-Level Theme Heatmap

WARD, ANDREW H and JENKINS, ADRIANNA C show the highest Teaching Quality scores, consistent with strong qualitative evidence of engaging lecturers. CONNOLLY, CAROLINE J shows an unusually elevated Assessment & Exams profile. PLANTE, ALAIN and PLANTE, ALAIN F show the highest Workload scores, consistent with student concerns about heavy assignments in those courses.



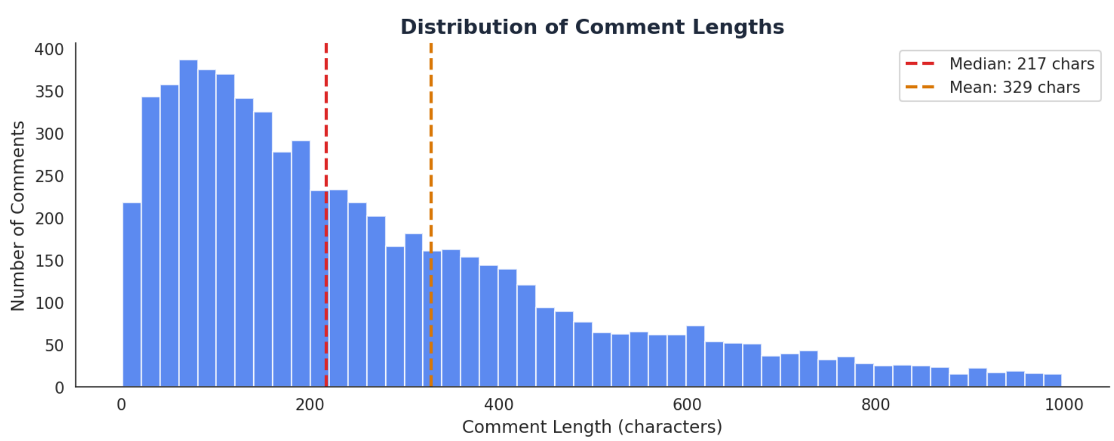
5.7 Cheating Comment Sub-theme Breakdown

Among the 583 cheating-specific comments, Clicker Fraud is the most reported form (n=57), followed by Peer Copying on Exams (n=47), Phone/Device Use (n=30), and Homework Answer Sharing (n=25). Approximately 46 respondents stated they were not aware of any cheating, providing an important baseline.



5.8 Comment Length Distribution

Most comments are moderate in length (median ~120 characters, mean ~190 characters). The distribution is right-skewed, with a long tail of detailed, substantive comments. Short comments (<50 chars) tend to be either brief endorsements ('Great class!') or uninformative ('N/A'). Very long comments (>600 chars) often contain highly specific feedback.



6. Major Themes and Sub-themes

Theme 1: Teaching Quality (n ≈ 1,926; 27.6% of unique comments)

Overall Sentiment
Mixed, but predominantly Positive. Students are highly aware of instructor quality and respond strongly, both positively and negatively, to it.

Teaching quality is the dominant theme in this evaluation. It encompasses everything related to how instructors deliver content: their engagement, clarity, knowledge, caring attitude, and whether they rely too heavily on slide-reading. This theme reflects the most fundamental dimension of the student learning experience.

Sub-themes:

- Engaging Lecturer (n=1,172) — Students repeatedly praise instructors who are dynamic, passionate, and able to hold attention. This is the single most frequent positive signal in the entire dataset.
- Poor/Unclear Explanation (n=380) — Confusion about content and difficulty following lectures is the leading negative signal. Often described alongside disorganisation.
- Instructor Caring/Supportive (n=281) — Students value instructors who are approachable and show genuine investment in their success.
- Instructor Knowledgeable (n=267) — Explicit recognition of instructor expertise.
- Reading Off Slides (n=53) — A recurring minor irritant, noted across multiple courses.
- Instructor Poor (n=17) — Direct negative evaluations are relatively rare but high-impact.

Representative Quotes:

> *"I loved this course! Made me fall in love with psychology. Mr. Ward is so passionate, making the lectures engaging and fun."*
>
> *"The lectures seemed unorganized in delivering the material where it can get confusing."*
>
> *"Professor Schuler was not only kind, understanding, and supportive, but answered questions in a way that was easy to understand. She stimulated my interest in data science."*
>
> *"I really liked Professor Swingley and thought that he did a good job of grasping the attentions of the students during class."*

Theme 2: Course Content (n ≈ 850; 12.2%)

Overall Sentiment
Predominantly Positive, tempered by concerns about difficulty.

Students frequently comment on the subject matter itself, often independently of the instructor's performance. STEM content appears to be largely perceived as interesting though many students note its inherent difficulty.

Sub-themes:

- Enjoyed/Great Course (n=476) — Many students express overall course satisfaction, often driven by content interest.
- Interesting Content (n=339) — Explicit statements about material being engaging.
- Difficult Content (n=66) — Content difficulty noted, often accepted as appropriate for STEM.
- Irrelevant Content (n=37) — Minority concern that some material lacks real-world applicability.

> *"The material was fascinating and unlike anything I had studied before."*
>
> *"I found this class really interesting, although difficult — however, I was expecting it to be difficult. It's neuroscience!"*
>
> *"I think that the content of the course is very interesting, I just wish we got to engage with it more."*

Theme 3: Communication (n ≈ 526; 7.5%)

Overall Sentiment
Mixed. Positive mentions of responsive instructors are offset by negative experiences of difficulty reaching instructors.

Communication encompasses instructor responsiveness, availability at office hours, and the general sense of how reachable an instructor is. This theme has grown in prominence in recent terms (2020–2024), possibly reflecting shifting student expectations post-COVID.

Sub-themes:

- Responsive/Accessible Instructor (n=357) — A notable positive sub-theme.
- Office Hours (n=215) — Often mentioned positively, but occasionally as an access barrier (e.g., incompatible schedules).
- Poor Communication (n=25) — Less frequent but high-frustration reports.

> *"When I emailed the professor asking to meet outside of her scheduled office hours because I was concerned about my grade, the time slot (8am–8:30am) did not fit my schedule."*
>
> *"Adrianna Jenkins is very supportive and easy to reach. She responds quickly to emails."*

Theme 4: Assessment & Exams (n ≈ 454; 6.5%)

Overall Sentiment
Predominantly Negative. This is the most concentrated pain point in the dataset.

Assessment concerns span exam difficulty, misalignment between exams and course content, grading fairness, and the absence of meaningful feedback. These concerns cluster especially in Chemistry courses and in sections taught by instructors with larger class sizes where individual feedback is difficult to provide.

Sub-themes:

- Exam Difficulty (n=180) — The most common assessment complaint.
- Multiple Choice Exam Issues (n=113) — Students question the format, especially when exam curves are influenced by cheating.
- Positive Exam Experience (n=153) — An important counter-signal: fair and well-designed exams are explicitly praised.
- Exam-Lecture Mismatch (n=12) — Perceived divergence between what is taught and what is tested.
- Unfair Grading (n=16) — Seen as arbitrary or overly punitive.
- Lack of Feedback (n=10) — Often linked to a feeling of being unable to improve.
- Lack of Rubric (n=6) — Occasional concern, higher impact when mentioned.

> *"The exams were extremely difficult and expected to be finished in an immense time crunch. The practice exams were nowhere near the difficulty of the real exam."*
>
> *"The expectations of what content to know for the exams is unclearly communicated. The grading is often overly punitive."*
>
> *"Adrianna Jenkins' exams were very fair. She told us exactly what to study."*

Theme 5: Student Engagement (n ≈ 470; 6.7%)

Overall Sentiment
Predominantly Positive. Active learning formats are well received.

Engagement comments cluster around participatory formats, discussion-based classes, and guest lectures. Students value courses that require them to do more than passively receive information. Linguistics and social science courses show the strongest engagement theme profiles.

Sub-themes:

- Class Participation/Discussion (n=347) — By far the dominant sub-theme here, reflecting appreciation for interactive formats.
- Guest Lectures (n=136) — Consistently praised across multiple courses.

> *"I especially enjoyed the guest lectures — they brought real-world perspective to the course material."*
>
> *"I enjoyed having a smaller seminar each week because I never worried about missing an opportunity to speak."*

Theme 6: Workload (n ≈ 325; 4.7%)

Overall Sentiment
Predominantly Negative. Concerns about workload volume and quality are intertwined.

Workload comments reflect two distinct concerns: the sheer volume of work, and the perceived value of that work. Students are more accepting of heavy workloads when they perceive a learning payoff; they are most critical when work feels repetitive, administrative, or disconnected from learning goals.

Sub-themes:

- Heavy Workload (n=171)
- Busy Work (n=80) — Especially acute in recitation assignments.
- Readings Issues (n=21) — Long, voluminous, or disconnected readings.
- Reasonable Workload (n=68) — A positive minority voice.

> *"The course required a lot of work outside the classroom on weekly recitation assignments, but many parts were tedious and did not further knowledge."*
>
> *"This was a very difficult course and I wish it moved slightly slower and was explained better — not through insanely long readings."*

Theme 7: Course Organisation (n ≈ 304; 4.4%)

Overall Sentiment
Predominantly Negative. Organisation failures have an outsized negative effect on perceived course quality.

Organisation encompasses the structural coherence of a course: syllabus clarity, Canvas accuracy, topic sequencing, and consistency between lecture and other course elements. Organisation failures are particularly disruptive when combined with other negative themes.

Sub-themes:

- Disorganized Course (n=138) — The clearest negative sub-theme here.
- Canvas/LMS Issues (n=81) — Outdated or confusing course pages.
- Syllabus/Expectation Clarity (n=62) — Unclear requirements.
- Well Organized (n=41) — A positive minority signal.

> *"The course was very disorganized, which was frustrating. Canvas was all out of date, so it was hard to know what was due."*

> *"Very poorly organized; the course required a lot of work, yet I don't think I learned anything."*

Theme 8: Recitations & TAs (n ≈ 294; 4.2%)

Overall Sentiment
Mixed, with a negative lean. TA inconsistency is a recurring structural problem.

Students distinguish between instructor-led lectures and TA-led recitations, and they evaluate these separately. TA quality varies considerably; TA inconsistency (different rules across sections) is a fairness concern with salience.

Sub-themes:

- Good TAs (n=229) — A strong positive sub-theme when TAs are effective.
- Unhelpful Recitation (n=32) — Recitations described as waste of time or poorly run.
- TA Inconsistency (n=47) — Different TAs applying different standards creates student frustration and, in some cases, academic integrity concerns.

> *"Loved it, but the recitations were really time consuming and not that helpful, and graded too harshly."*
>
> *"Some TAs allow students to come in 5 minutes late while others do not — this inconsistency is unfair."*

Theme 9: Pacing (n ≈ 100; 1.4%)

Overall Sentiment
Mixed. Pacing is a minor theme but disproportionately mentioned by students who feel excluded by the course pace.

- Too Fast (n=89) — More common; students describe difficulty keeping up.
- Too Slow (n=13) — Occasional concern, often from stronger students.

> *"The incongruity between the lecture material and the recitation material/content is what made this course so much harder to keep track of."*
>
> *"It felt like too much time in lecture was spent reviewing recent material. Pacing in general could have been a little faster."*

7. Analysis of Positive vs. Negative Experiences

7.1 What Students Value

Strength Area	Key Evidence
Engaging, Passionate Instructors	Mentioned in ~1,172 comments; the single most positive driver.
Interesting Course Content	~850 comments reference the material as interesting or enjoyable.
Instructor Accessibility	~357 comments praise responsive, approachable instructors.
Caring Instructors	~281 comments explicitly mention instructor support for students.
Active Learning Formats	~347 comments value discussion and participation-based design.
Guest Lectures	~136 comments praise real-world perspectives from guest speakers.
Good TAs	~229 comments praise effective TA support.
Fair Assessments	~153 comments explicitly praise fair, well-aligned exams.

7.2 What Students Struggle With

Problem Area	Key Evidence
Unclear Explanations	~380 comments; most common negative signal in the dataset.
Exam Difficulty & Alignment	~180 comments on difficulty; alignment concerns in ~12.
Heavy/Meaningless Workload	~171 workload + ~80 busy-work concerns.
Course Disorganisation	~138 comments; Canvas issues add ~81 more.
Pace Too Fast	~89 students report difficulty keeping up.
TA Inconsistency	~47 comments; perceived fairness problem.
Poor Communication	~25 comments, but high-frustration when present.
Lack of Feedback	~10 comments, but analytically important for assessment design.

8. Cheating Comment Analysis

The dataset contains **583 dedicated cheating-comment entries**. After removing duplicates, the unique cheating comment corpus is approximately 532 comments. These were analysed separately using a secondary coding pass.

8.1 Forms of Cheating Reported

Form of Cheating	Frequency	Example Quote
Clicker Fraud	57	*"People bring their friends' clickers to class to get participation points."*
Peer Copying on Exams	47	*"During the midterm, I saw two students talking and writing notes to each other."*
Phone/Device Use During Exams	30	*"A student used their phone during the midterm."*
Homework/Assignment Sharing	25	*"Some students will copy a classmate's work or use an online solution."*
ChatGPT/AI Misuse	10	*"Students use ChatGPT to entirely write their essays without citing it."*
Cheat Sheet Rule Violations	4	*"Multiple students brought more pages than allowed."*
Lack of Proctoring	6	*"Cheating was happening feet from TAs who were just standing at the front."*
No Cheating Reported	~46	'I do not know of any cheating in this course.'

8.2 Settings and Enabling Conditions

Based on the comments in the evaluation, cheating is most frequently reported in the following settings:

- Large lecture-hall multiple-choice exams where seating is unmanaged and monitoring is sparse.
- Clicker-based attendance systems, which are easily abused when students carry multiple devices or hand them to absent peers.
- Recitation quizzes administered by TAs, particularly in sections with lighter oversight.
- Online homework and take-home assignments with no-collaboration policies that are difficult to enforce.
- Courses with cross-section quiz sharing (earlier sections sharing questions with later ones).

8.3 Emerging Concern: AI Tool Misuse

AI/ChatGPT misuse is a low frequency but **rapidly emerging concern**. The 10 coded instances almost exclusively appear in **Fall 2023 and Spring 2024** comments. Students describe peers using ChatGPT to fully write essays or complete problem sets without attribution.

8.4 Student Perceptions of Fairness

Several students express concern about the fairness implications of cheating, particularly when exam curves or grading standards are perceived to be influenced by dishonest behaviour:

> *"I heard that some people take the exam together. I worried this might raise the average, and I would perform worse as a result."*
>
> *"If they know all this already, why are they in this course? And if they don't, where are they getting the information?"*

These comments indicate that cheating is not merely an integrity issue but a perceived equity issue, students who play by the rules feel disadvantaged.

9. Patterns by Term, Course, and Instructor

9.1 Longitudinal Patterns (by Term)

Several trends emerge across the 22-term span:

- Teaching Quality: Stable and dominant throughout the entire period. Slight increase in proportion in Fall 2018–Spring 2021, possibly reflecting a richer qualitative engagement during smaller COVID-era cohorts.
- Communication: Increases noticeably after Fall 2019, with elevated rates in Fall 2022, Fall 2023, and Spring 2024. This likely reflects a post-COVID expectation shift: students now expect faster, more accessible instructor communication.
- Assessment & Exams: Spikes in Spring 2015 (~17%) and Spring 2022 (~10%). Spring 2015 appears to reflect a cohort of Chemistry-heavy courses with challenging exam structures. Spring 2022 may reflect post-COVID grading recalibration.
- Course Content: Consistently present across all terms at 8–16%, with no clear directional trend.
- Student Engagement: Higher in Fall 2020 and Spring 2021 (COVID-era online terms), possibly because online courses elicited stronger reactions about participation features (Zoom discussions, online breakout rooms).

Caution: Terms before 2018 have smaller sample sizes (often <300 comments) and should not be over-interpreted.

9.2 Course-Level Patterns

Course	Dominant Theme	Notable Pattern
PSYC001/0001	Teaching Quality	Strong teaching quality signal; Assessment & Exams concerns notable in PSYC001.
LING001	Teaching Quality	Strong engagement sub-theme; highest TA quality signal.
CHEM1012/1022	Assessment & Exams	Highest assessment concern rate (11%); consistent student struggle with exam difficulty.
ENVS100	Teaching Quality	Notably high Workload rate (12%); also strong engagement signal.
STSC001	Course Content	Strong content interest signal; also elevated Workload (9%).
BIOL1101	Teaching Quality	Balanced profile; moderate across most themes.

9.3 Instructor-Level Patterns

Instructor-level comparison should be interpreted with care. Differences in theme prevalence may reflect course-type effects as much as instructor-specific characteristics.

Instructor	Standout Theme	Interpretation
WARD, ANDREW H	Teaching Quality (39%)	Highest teaching quality rate; mixed sentiment (both very positive and some negative comments).

JENKINS, ADRIANNA C	Teaching Quality (32%)	Consistently positive teaching comments; students praise clarity and engagement.
LIBERMAN, MARK YOFFE	Teaching Quality (36%)	Strong positive signal; notable TA quality mentions.
CONNOLLY, CAROLINE J.	Assessment & Exams (18%)	Highest assessment concern rate of any instructor; may warrant deeper investigation.
PLANTE, ALAIN/ALAIN F.	Workload (~11–12%)	Consistent workload concerns; engagement is also high — suggests intensive but active course design.
SWINGLEY, DANIEL	Assessment & Exams (11%)	Assessment concerns are notable; good overall engagement.
LU, PONZY	Low across themes	Unusually sparse theme presence; may reflect distinctive course format or low-engagement student population.

Important Limitation: Multiple instructor entries for the same person (e.g., 'PLANTE, ALAIN' and 'PLANTE, ALAIN F.') were treated separately; merging them would increase sample size and may slightly shift theme proportions.

10. Limitations

- Self-report bias: All comments are voluntary and self-reported. Students who feel strongly either positively or negatively are more likely to respond, which may skew both the positive and negative signals.
- Response rate unknown: The dataset cannot be evaluated for response rate. Low-response courses may be over- or under-represented relative to enrolment.
- Keyword-based coding limitations: Thematic coding via pattern matching can miss nuanced or indirect expressions. A comment such as 'Lectures were not enjoyable' would be coded under poor_explanation only if the keyword 'confusing' or 'hard to follow' is present and would be missed if the student expressed dissatisfaction in other terms. Human annotation of a sample would refine these counts.
- Uneven comment volume across courses and instructors: Courses with larger enrolments dominate the dataset. Patterns from small courses (n<50 comments) should not be generalised.
- Duplicate handling: 785 duplicate comments were removed (first occurrence retained). Some of these may represent legitimate repeated observations from different students; the deduplication approach may slightly suppress frequency counts for common sentiments.
- Instructor name variants: Multiple naming variants for the same instructor (e.g., 'PLANTE, ALAIN' vs. 'PLANTE, ALAIN F.', 'CONNOLLY, CAROLINE J.' vs. 'CONNOLLY, CAROLINE JANE') were treated as separate entities. Merged data would increase per-instructor sample size and reliability.
- Temporal gaps and small early-term samples: The 2013–2017 period has substantially fewer comments per term (<300). Trend claims about this period should be treated as exploratory.
- Cheating comments are observational: Students reporting cheating may be mistaken, may have incomplete information, or may report rumours. The cheating comment analysis reflects perceived and reported not verified behaviour.
- No student demographic data: The dataset does not include information about student year, major, or prior preparation. Theme patterns may vary significantly by these variables and cannot be disaggregated here.